

Overview

Background



Drug-Drug Interaction Risk

Multiple medications can increase the risk of drug-drug interactions (DDIs).



Limited Biomedical Context

DDI-only graphs do not include drug-gene/protein and drug-disease relationships.



Core Research Question

Can richer biomedical knowledge improve DDI prediction?

Goals



Biomedical Knowledge Comparison

Test whether gene/protein and disease relationships improve DDI prediction.



Robustness & Generalization

Evaluate consistency across repeated and challenging test settings.



CHEERS Platform Development

Build a source-grounded platform for medicines, diseases, interactions, and evidence.

Methodology

Model Development & Evaluation

PrimeKG Graph Compositions



G0- DDI only
G1- +Drug-Gene/Protein
G2- + Drug-Disease
G3- +Both

R-GCN Model



- 128-D embeddings
- 2-layer R-GCN
- ReLU + Dropout 0.2
- DistMult decoder

Evaluation and Robustness



- Relation-level analysis
- Five-seed analysis
- Cold-start evaluation
- Structural / Hybrid R-GCN
- DDInter external evaluation
- Classification analysis

Trained G3 Seed-44



- Selected trained G3 model
- Precomputed embeddings
- Exported NumPy runtime
- Used by web application

Model Inference (Web Application)



- Load frozen NumPy arrays
- Calculate DistMult scores
- Rank candidate drug pairs
- Return ranked predictions
- No retraining, no GPU required

Results

Five-seed mean results

Graph	Composition	MRR	Hits@1	Hits@5	Hits@10	Accuracy	Precision	Recall	F1
G0	DDI only	0.5273	0.4805	0.5727	0.6094	0.9192	0.9354	0.9005	0.9176
G1	+ Drug-Gene/Protein	0.5310	0.4842	0.5754	0.6127	0.9219	0.9368	0.9048	0.9205
G2	+ Drug-Disease	0.5268	0.4802	0.5717	0.6086	0.9199	0.9353	0.9022	0.9185
G3	+ Both	0.5342	0.4863	0.5805	0.6181	0.9227	0.9373	0.9061	0.9214



0.5342
Highest mean MRR (G3)

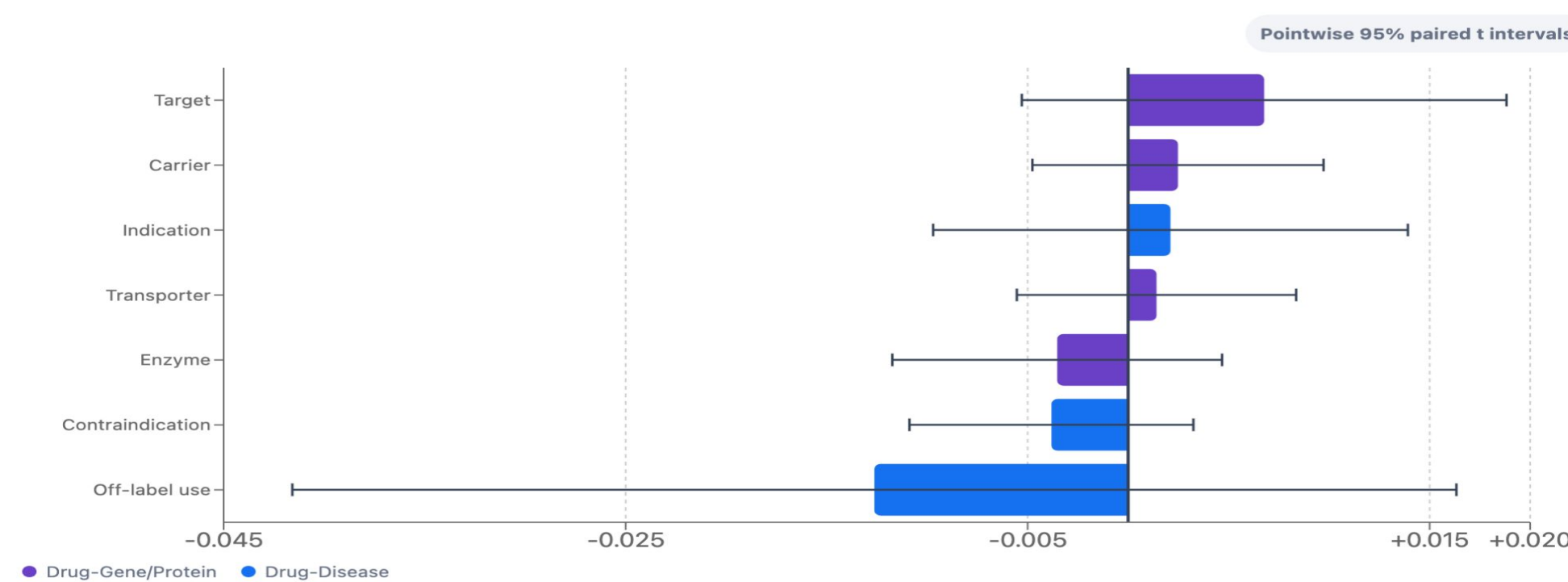


5/5
Seeds improved over G0



92.27%
Secondary Classification Accuracy 0.9214 F1

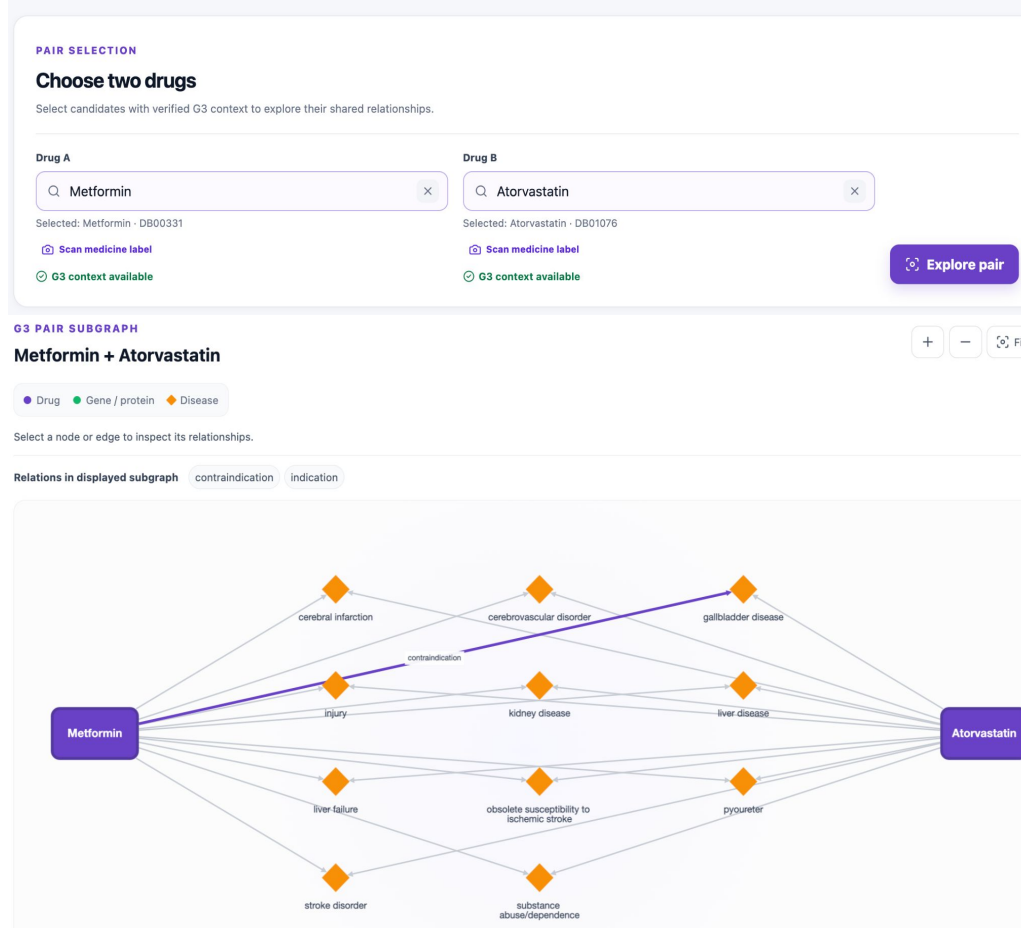
Mean Δ MRR = +0.0069; exact two-sided sign-flip $p = 0.0625$.

Mean Δ MRR relative to G0

- Target (Drug-Protein) shows the highest mean MRR.
- External Evaluation: DDInter was used for exploratory transfer testing.

Graph Explorer

Explore shared gene/protein and disease associations available in the verified G3 knowledge graph. This context is not a causal explanation of a model score and does not establish clinical safety or harm.

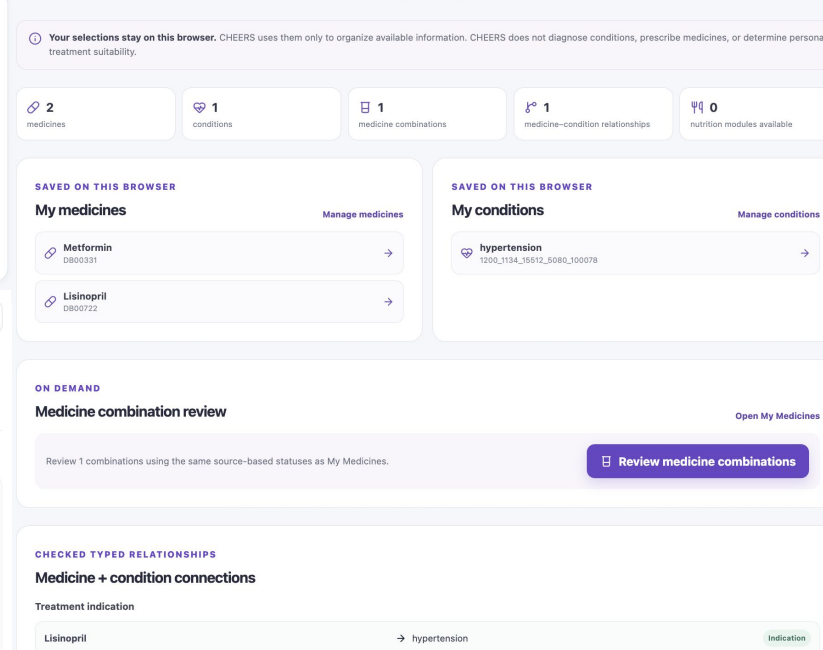


G3 Link Ranking & Subgraph

Ranked DDI candidates with 1-hop biomedical graph context.

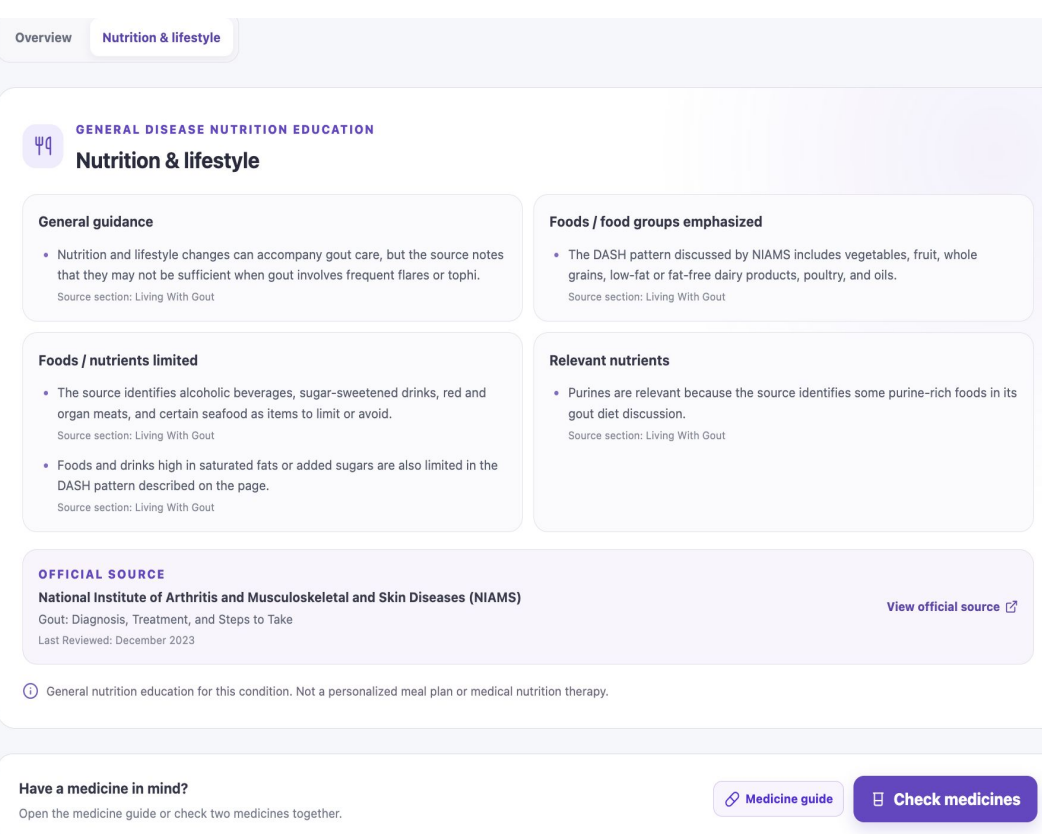
Your selected medicines and conditions, brought together in one place.

Use this overview to reach available CHEERS information without creating a clinical profile.



My Medicines & My Health

Personalized overview based on saved medicines and conditions.



Source-Grounded Health Information

Combines openFDA drug-label information, PubMed literature, and reviewed NIH nutrition and lifestyle sources.

Applications



Medicine Information

Integrates medicine, disease, interaction, and lifestyle information in one interface.



Evidence Exploration

Connects PrimeKG graph context with openFDA and PubMed literature.



Research & Education

Provides access to DDI predictions, biomedical relationships, experimental results, and their limitations.